

```
1  =====
2
3  Q1. Student Information
4
5  Create a student object containing:
6
7  - id
8  - name
9  - age
10 - course
11 - city
12
13 Write a function displayStudent(student).
14
15 Use object destructuring inside the function to print all details.
16
17
18 -----
19
20 Q2. Product Discount Calculator
21
22 Create a product object containing:
23
24 - name
25 - price
26 - discount
27
28 Write a function calculateFinalPrice(product).
29
30 Calculate:
31 - Discount Amount
32 - Final Price
33
34 Use object destructuring inside the function.
35
36
37 -----
38
39 Q3. Employee List
40
41 Create an array of employee objects.
42
43 Each employee contains:
44
45 - name
46 - salary
47
48 Using a loop, print:
49
50 Employee Name : _____
51 Salary : _____
52
53 -----
54
55 Q4. Number Checker
```

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56
57 Create an object containing:
58
59 - number
60
61 Write a function checkNumber(obj).
62
63 Use object destructuring inside the function.
64
65 Print whether the number is:
66
67 - Even
68 - Odd
69
70 -----
71
72 Q5. Find a Student
73
74 Create an array of student objects.
75
76 Each object contains:
77
78 - id
79 - name
80 - age
81
82 Take an ID from the user.
83
84 Use the find() method to search for the student.
85
86 If found, print all details.
87
88 Otherwise print:
89
90 Student Not Found
91
92 =====
93
94 Q6. Highest Marks
95
96 Create an array of student objects.
97
98 Each object contains:
99
100 - name
101 - marks
102
103 Using a loop,
104
105 find the student who scored the highest marks.
106
107 Use object destructuring only while printing the topper.
108
109 -----
110
```

```
111 Q7. Count Adults
112
113 Create an array of person objects.
114
115 Each object contains:
116
117 - name
118 - age
119
120 Use filter() to create a new array containing only adults
121 (age >= 18).
122
123 Print:
124
125 - Adult count
126 - Adult names
127
128 -----
129
130 Q8. Shopping Cart
131
132 Create an array of product objects.
133
134 Each object contains:
135
136 - name
137 - price
138 - quantity
139
140 Use reduce() to calculate the total bill.
141
142 Create a function calculateBill(products).
143
144 Print the final amount.
145
146
147 -----
148
149 Q9. Search Product
150
151 Create an array of product objects.
152
153 Each product contains:
154
155 - name
156 - price
157 - category
158
159 Take a product name from the user.
160
161 Use find() to search for the product.
162
163 If found,
164
165 use object destructuring to print:
```

```
166
167 - Name
168 - Price
169 - Category
170
171 Otherwise print:
172
173 Product Not Found
174
175
176 -----
177
178 Q10. Student Report
179
180 Create an array of student objects.
181
182 Each object contains:
183
184 - name
185 - marks
186
187 Use map() to create a new array containing:
188
189 {
190     name,
191     grade
192 }
193
194 Grade Rules
195
196 90+ -> A
197 75+ -> B
198 60+ -> C
199 Below 60 -> D
200
201 Print the new array.
202
203
204 =====
205
206 Q11. Department Salary Report
207
208 Create an array of employee objects.
209
210 Each employee contains:
211
212 - id
213 - name
214 - department
215 - salary
216
217 Write a function:
218
219 departmentSalary(employees)
220
```

221 Use reduce() to calculate the total salary paid in each department.

222

223 Example Output

224

225 IT : 120000

226 HR : 75000

227 Sales : 90000

228

229

230 -----

231

232 Q12. Student Result Analysis

233

234 Create an array of student objects.

235

236 Each student contains:

237

238 - name

239 - marks (array of 3 subjects)

240

241 Write a function analyzeStudents(students).

242

243 For every student:

244

245 - Calculate Total

246 - Calculate Average

247 - Print Pass/Fail

248

249 Use array destructuring for marks.

250

251 Use object destructuring wherever appropriate.

252

253

254 -----

255

256 Q13. Library Management

257

258 Create an array of book objects.

259

260 Each book contains:

261

262 - title

263 - author

264 - price

265 - available

266

267 Tasks

268

269 1. Use filter() to display only available books.

270

271 2. Use map() to print only:

272

273 - Title

274 - Author

275

```
276 Use object destructuring inside map().
277
278
279 -----
280
281 Q14. Product Inventory
282
283 Create an array of product objects.
284
285 Each product contains:
286
287 - id
288 - name
289 - stock
290 - price
291
292 Write separate functions to:
293
294 1. Display all products.
295
296 2. Find a product using find().
297
298 3. Check if every product is in stock using every().
299
300 4. Check if some products are out of stock using some().
301
302 5. Calculate total inventory value using reduce().
303
304 Use object destructuring only while printing product details.
305
306
307 -----
308
309 Q15. Student Management Report
310
311 Create an array of student objects.
312
313 Each student contains:
314
315 - id
316 - name
317 - city
318 - course
319 - marks (array)
320
321 Write separate functions:
322
323 1. displayStudents()
324
325 2. findTopper()
326
327 3. searchStudent(id)
328
329 4. calculateAverage()
330
```

```
331 5. passedStudents()
332
333 Requirements
334
335 • Use map() to create a report card.
336 • Use filter() to display passed students.
337 • Use reduce() to calculate the class average.
338 • Use find() to search by ID.
339 • Use object destructuring whenever printing student details.
340 • Use array destructuring while accessing marks.
341
342
343 =====
344 BONUS CHALLENGE
345 =====
346
347 Create a complete Student Management System.
348
349 Each student object contains:
350
351 - id
352 - name
353 - age
354 - city
355 - course
356 - marks (array of 3 subjects)
357
358 Write separate functions for:
359
360 1. Display all students.
361 2. Search student by ID (find()).
362 3. Search students by city (filter()).
363 4. Display students by course (filter()).
364 5. Find the topper.
365 6. Find the lowest scorer.
366 7. Calculate total marks of every student (map()).
367 8. Calculate average marks of every student (map()).
368 9. Display only passed students (filter()).
369 10. Display only failed students (filter()).
370 11. Count total passed students.
371 12. Count total failed students.
372 13. Check if every student has passed (every()).
373 14. Check if at least one student scored above 95 (some()).
374 15. Calculate class average using reduce().
375 16. Calculate highest marks.
376 17. Calculate lowest marks.
377 18. Sort students by name.
378 19. Sort students by average marks.
379 20. Print a complete report card for every student.
380
381 Use:
382 ✓ Objects
383 ✓ Object Destructuring
384 ✓ Array Destructuring
385 ✓ Functions
```

```
386  ✓ Loops
387  ✓ map()
388  ✓ filter()
389  ✓ reduce()
390  ✓ find()
391  ✓ some()
392  ✓ every()
393
394  wherever they are most appropriate—not in every question.
395
396  =====
397  END OF PRACTICE SET
398  =====
```